

# A combined Liouville integrable hierarchy associated with a fourth-order matrix spectral problem

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## Abstract

This paper aims to propose a fourth-order matrix spectral problem involving four potentials and generate an associated Liouville integrable hierarchy via the zero curvature formulation. A bi-Hamiltonian formulation is furnished by applying the trace identity and a recursion operator is explicitly worked out, which exhibits the Liouville integrability of each model in the resulting hierarchy. Two specific examples, consisting of novel generalized combined nonlinear Schrödinger equations and modified Korteweg–de Vries equations, are given.

**Keywords:** integrable hierarchy, Hamiltonian structure, zero curvature equation, Lax pair, matrix spectral problem

## 1. Introduction

Lax pairs of matrix spectral problems [1] are the key in generating integrable models. Based on matrix spectral problems, ones can explore remarkable integrable properties, for example, infinitely many symmetries and conserved quantities [2, 3]. There are diverse applications of integrable models in physical and engineering sciences, particularly in nonlinear optics, water waves and quantum mechanics.

Among typical examples of integrable models are the Ablowitz–Kaup–Newell–Segur hierarchy [4] and its various hierarchies of integrable couplings [6]. Matrix Lie algebras play a crucial role in formulating meaningful Lax pairs [5, 7], based on which one can identify and classify integrable models into specific categories. There are many examples with one or two components but few examples with multi-components. In this paper, we would like to propose a new matrix spectral problem based on a specific matrix Lie algebra and compute an associated integrable hierarchy.

Let us recall the zero curvature formulation for constructing integrable hierarchies briefly (see [7, 8] for details).

In our discussion, we denote a  $q$ -dimensional column potential vector by  $u = (u_1, \dots, u_q)^T$  and the spectral parameter by  $\lambda$ . First, take a given loop matrix algebra  $\tilde{g}$  with the loop parameter  $\lambda$ , and formulate a spatial spectral matrix:

$$\mathcal{M} = \mathcal{M}(u, \lambda) = u_1 F_1(\lambda) + \dots + u_q F_q(\lambda) + F_0(\lambda), \quad (1.1)$$

where the elements  $F_1, \dots, F_q$  are linear independent in  $\tilde{g}$ . We assume that the above element  $F_0$  is pseudo-regular:

$$\text{Im ad}_{F_0} \oplus \text{Ker ad}_{F_0} = \tilde{g}, [\text{Ker ad}_{F_0}, \text{Ker ad}_{F_0}] = 0, \quad (1.2)$$

where  $\text{ad}_{F_0}$  denotes the adjoint action of  $F_0$  on the Lie algebra  $\tilde{g}$ . Under this condition, there is a Laurent series solution  $Y = \sum_{n \geq 0} \lambda^{-n} Y^{[n]}$  to stationary zero curvature equation

$$Y_x = [\mathcal{M}, Y] \quad (1.3)$$

in the underlying loop algebra  $\tilde{g}$ .

Second, we set an infinite sequence of temporal spectral matrices

$$\mathcal{N}^{[m]} = (\lambda^m Y)_+ + \Delta_r = \sum_{n=0}^m \lambda^{m-n} Y^{[n]} + \Delta_m, \quad m \geq 0, \quad (1.4)$$

where  $\Delta_m \in \tilde{g}$ ,  $m \geq 0$ , as the other parts of a sequence of Lax pairs, to generate a hierarchy of integrable models:

$$u_{t_m} = X^{[m]} = X^{[m]}(u), \quad m \geq 0, \quad (1.5)$$

via the zero curvature equations:

$$\mathcal{M}_{t_m} - \mathcal{N}_x^{[m]} + [\mathcal{M}, \mathcal{N}^{[m]}] = 0, \quad m \geq 0. \quad (1.6)$$

These equations represent the solvability conditions of the spatial and temporal matrix spectral problems:

$$\varphi_x = \mathcal{M}\varphi, \quad \varphi_{t_m} = \mathcal{N}^{[m]}\varphi, \quad m \geq 0. \quad (1.7)$$

Finally, we construct Hamiltonian formulations by the so-called trace identity:

$$\frac{\delta}{\delta u} \int \text{tr} \left( Y \frac{\partial \mathcal{M}}{\partial \lambda} \right) dx = \lambda^{-\kappa} \frac{\partial}{\partial \lambda} \lambda^{\kappa} \text{tr} \left( Y \frac{\partial \mathcal{M}}{\partial u} \right), \quad (1.8)$$

where  $\frac{\delta}{\delta u}$  is the variational derivative with respect to  $u$ , and  $\kappa$  is a constant, independent of  $\lambda$ , for the hierarchy (1.5). Together with a recursion operator  $\Phi$  determined by  $X^{[m+1]} = \Phi X^{[m]}$ , this shows the Liouville integrability (see, e.g. [7, 9]) of the hierarchy (1.5).

Many hierarchies of Liouville integrable models have been generated via the zero curvature formulation (see, e.g.

set  $\tilde{g}$  of block matrices

$$\tilde{g} = \left\{ A = \begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix}_{2r \times 2r} \mid A_4 = TA_1T^{-1}, A_3 = \delta TA_2T^{-1} \right\}. \quad (2.2)$$

It is easy to see this forms a matrix Lie algebra under the matrix commutator  $[A, B] = AB - BA$ . We will use this Lie algebra with  $r = 2$  and

$$T = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \text{ or } \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad (2.3)$$

to formulate a specific spectral matrix below.

Let  $\alpha_1$  and  $\alpha_2$  be two arbitrary constants, and  $u = u(x, t) = (u_1, u_2, u_3, u_4)^T$  ( $x, t \in \mathbb{R}$ ), a four-component potential vector. Assume that

$$\alpha = \alpha_1 + \alpha_2 \neq 0, \quad \delta \neq 0. \quad (2.4)$$

Motivated by recent studies on matrix spectral problems with four potentials (see, e.g. [23–25] and [26, 27]), let us introduce a matrix spectral problem of the form:

$$\varphi_x = \mathcal{M}\varphi = \mathcal{M}(u, \lambda)\varphi, \quad \mathcal{M} = \begin{bmatrix} 0 & u_1 & u_2 & \alpha_1\lambda \\ u_3 & 0 & \alpha_2\lambda & u_4 \\ \delta u_4 & -\delta\alpha_2\lambda & 0 & -u_3 \\ -\delta\alpha_1\lambda & \delta u_2 & -u_1 & 0 \end{bmatrix}, \quad (2.5)$$

[4–19]). When  $q = 2$ , e.g. in the case of two components, we have the Ablowitz–Kaup–Newell–Segur hierarchy [4], the Heisenberg hierarchy [20], the Kaup–Newell hierarchy [21] and the Wadati–Konno–Ichikawa hierarchy [22]. All of the corresponding spectral matrices are  $2 \times 2$  and involve two potentials.

In this paper, we aim to present a hierarchy of four-component Liouville integrable models through the zero curvature formulation. The starting point is a specific matrix spectral problem. The corresponding bi-Hamiltonian formulation is furnished via an application of the trace identity. Two illustrative examples of generalized combined integrable nonlinear Schrödinger and modified Korteweg–de Vries models are worked out. The last section provides a conclusion and a few concluding remarks.

## 2. A matrix spectral problem and its four-component integrable hierarchy

Let  $\delta$  be an arbitrary number, and  $T$  be a square matrix of order  $r$  such that

$$T^2 = -I_r, \quad (2.1)$$

where  $I_r$  denotes the identity matrix of order  $r$ . Let us form a

where  $\lambda$  is again the spectral parameter. This spectral matrix is from the matrix Lie algebra previously defined, with  $r = 2$  and  $T$  by (2.3). The spectral problem is not any reduction of the matrix Ablowitz–Kaup–Newell–Segur spectral problem (see, e.g. [28]), but it still generates an integrable hierarchy, each of which is bi-Hamiltonian and possesses combined structures.

As usual, to generate an associated Liouville integrable hierarchy, we first solve the corresponding stationary zero curvature equation (1.3). The solution  $Y$  is assumed to be of the form:

$$Y = \begin{bmatrix} a & b & e & f \\ c & -a & f & g \\ \delta g & -\delta f & -a & -c \\ -\delta f & \delta e & -b & a \end{bmatrix} = \sum_{n \geq 0} \lambda^{-n} Y^{[n]}, \quad (2.6)$$

where all basic objects are assumed to be of Laurent series type:

$$\begin{cases} a = \sum_{n \geq 0} \lambda^{-n} a^{[n]}, & b = \sum_{n \geq 0} \lambda^{-n} b^{[n]}, & c = \sum_{n \geq 0} \lambda^{-n} c^{[n]}, \\ e = \sum_{n \geq 0} \lambda^{-n} e^{[n]}, & f = \sum_{n \geq 0} \lambda^{-n} f^{[n]}, & g = \sum_{n \geq 0} \lambda^{-n} g^{[n]}. \end{cases} \quad (2.7)$$

We take a solution of the above form, because this is the form that the commutator between any matrix in  $\tilde{g}$  and the spectral matrix  $\mathcal{M}$  takes. Clearly, the corresponding stationary zero

curvature equation (1.3) reads

$$\begin{cases} a_x = cu_1 + \delta gu_2 - bu_3 - \delta eu_4, \\ b_x = \alpha \delta \lambda e - 2au_1 - 2\delta fu_2, \\ c_x = \alpha \delta \lambda g + 2au_3 - 2\delta fu_4, \end{cases} \quad (2.8)$$

$$\begin{cases} e_x = -\alpha \lambda b - 2au_2 + 2fu_1, \\ g_x = -\alpha \lambda c + 2au_4 + 2fu_3, \\ f_x = gu_1 - cu_2 + eu_3 - bu_4. \end{cases} \quad (2.9)$$

These equations equivalently yield the initial conditions:

$$a_x^{[0]} = 0, b^{[0]} = c^{[0]} = e^{[0]} = g^{[0]} = 0, f_x^{[0]} = 0, \quad (2.10)$$

and the recursion relations to determine the Laurent series solution  $Y$ :

$$\begin{cases} b^{[n+1]} = \frac{1}{\alpha}(-e_x^{[n]} - 2a^{[n]}u_2 + 2f^{[n]}u_1), \\ c^{[n+1]} = \frac{1}{\alpha}(-g_x^{[n]} + 2a^{[n]}u_4 + 2f^{[n]}u_3), \end{cases} \quad (2.11)$$

$$\begin{cases} e^{[n+1]} = \frac{1}{\alpha \delta}(b_x^{[n]} + 2\delta f^{[n]}u_2 + 2a^{[n]}u_1), \\ g^{[n+1]} = \frac{1}{\alpha \delta}(c_x^{[n]} + 2\delta f^{[n]}u_4 - 2a^{[n]}u_3), \end{cases} \quad (2.12)$$

$$\begin{cases} a_x^{[n+1]} = c^{[n+1]}u_1 + \delta g^{[n+1]}u_2 - b^{[n+1]}u_3 - \delta e^{[n+1]}u_4, \\ f_x^{[n+1]} = g^{[n+1]}u_1 - c^{[n+1]}u_2 + e^{[n+1]}u_3 - b^{[n+1]}u_4, \end{cases} \quad (2.13)$$

where  $n \geq 0$ . To achieve the uniqueness of Laurent series solutions, we take the initial data,

$$a^{[0]} = \frac{1}{2}\beta, f^{[0]} = \frac{1}{2}\gamma, \quad (2.14)$$

where  $\beta$  and  $\gamma$  are two arbitrary constants, and assume the constants of integration to be zero,

$$a^{[n]}|_{u=0} = 0, f^{[n]}|_{u=0} = 0, n \geq 1. \quad (2.15)$$

In this way, one can work out that

$$\begin{cases} b^{[1]} = \frac{1}{\alpha}(\gamma u_1 - \beta u_2), c^{[1]} = \frac{1}{\alpha}(\gamma u_3 + \beta u_4), \\ e^{[1]} = \frac{1}{\alpha \delta}(\beta u_1 + \delta \gamma u_2), g^{[1]} = \frac{1}{\alpha \delta}(-\beta u_3 + \delta \gamma u_4), \\ a^{[1]} = f^{[1]} = 0; \\ b^{[2]} = -\frac{1}{\alpha^2 \delta}(\beta u_{1,x} + \delta \gamma u_{2,x}), c^{[2]} = \frac{1}{\alpha^2 \delta}(\beta u_{3,x} - \delta \gamma u_{4,x}), \\ e^{[2]} = \frac{1}{\alpha^2 \delta}(\gamma u_{1,x} - \beta u_{2,x}), g^{[2]} = \frac{1}{\alpha^2 \delta}(\gamma u_{3,x} + \beta u_{4,x}), \\ a^{[2]} = \frac{1}{\alpha^2 \delta}[(\beta u_3 - \delta \gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2], \\ f^{[2]} = \frac{1}{\alpha^2 \delta}[(\gamma u_1 - \beta u_2)u_3 + (\beta u_1 + \delta \gamma u_2)u_4]; \\ b^{[3]} = \frac{1}{\alpha^3 \delta}[-\gamma u_{1,xx} + \beta u_{2,xx} + 2(\gamma u_3 + \beta u_4)(u_1^2 - \delta u_2^2) - 4(\beta u_3 - \delta \gamma u_4)u_1 u_2], \\ c^{[3]} = \frac{1}{\alpha^3 \delta}[-\gamma u_{3,xx} - \beta u_{4,xx} + 2(\gamma u_1 - \beta u_2)(u_3^2 - \delta u_4^2) + 4(\beta u_1 + \delta \gamma u_2)u_3 u_4], \end{cases}$$

$$\begin{cases} e^{[3]} = \frac{1}{\alpha^3 \delta^2} [-\beta u_{1,xx} - \delta \gamma u_{2,xx} + 2(\beta u_3 - \delta \gamma u_4)(u_1^2 - \delta u_2^2) + 4\delta(\gamma u_3 + \beta u_4)u_1 u_2], \\ g^{[3]} = \frac{1}{\alpha^3 \delta^2} [\beta u_{3,xx} - \delta \gamma u_{4,xx} - 2(\beta u_1 + \delta \gamma u_2)(u_3^2 - \delta u_4^2) + 4\delta(\gamma u_1 - \beta u_2)u_3 u_4], \\ a^{[3]} = \frac{1}{\alpha^3 \delta} [(\gamma u_3 + \beta u_4)u_{1,x} - (\beta u_3 - \delta \gamma u_4)u_{2,x} - (\gamma u_1 - \beta u_2)u_{3,x} - (\beta u_1 + \delta \gamma u_2)u_{4,x}], \\ f^{[3]} = \frac{1}{\alpha^3 \delta^2} [-(\beta u_3 - \delta \gamma u_4)u_{1,x} - \delta(\gamma u_3 + \beta u_4)u_{2,x} + (\beta u_1 + \delta \gamma u_2)u_{3,x} - \delta(\gamma u_1 - \beta u_2)u_{4,x}]; \end{cases}$$

and

$$\begin{cases} b^{[4]} = \frac{1}{\alpha^4 \delta^2} \{ \beta u_{1,xxx} + \delta \gamma u_{2,xxx} - 6[(\beta u_3 - \delta \gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{1,x} \\ \quad - 6\delta[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta \gamma u_4)u_2]u_{2,x} \}, \\ c^{[4]} = \frac{1}{\alpha^4 \delta^2} \{ -\beta u_{3,xxx} + \delta \gamma u_{4,xxx} + 6[(\beta u_3 - \delta \gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{3,x} \\ \quad - 6\delta[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta \gamma u_4)u_2]u_{4,x} \}, \\ e^{[4]} = \frac{1}{\alpha^4 \delta^2} \{ -\gamma u_{1,xxx} + \beta u_{2,xxx} + 6[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta \gamma u_4)u_2]u_{1,x} \\ \quad - 6[(\beta u_3 - \delta \gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{2,x} \}, \\ g^{[4]} = \frac{1}{\alpha^4 \delta^2} \{ -\gamma u_{3,xxx} - \beta u_{4,xxx} + 6[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta \gamma u_4)u_2]u_{3,x} \\ \quad + 6[(\beta u_3 - \delta \gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{4,x} \}, \\ a^{[4]} = \frac{1}{\alpha^4 \delta^2} [-(\beta u_3 - \delta \gamma u_4)u_{1,xx} - \delta(\gamma u_3 + \beta u_4)u_{2,xx} - (\beta u_1 + \delta \gamma u_2)u_{3,xx} + \delta(\gamma u_1 - \beta u_2)u_{4,xx} \\ \quad + (\beta u_{3,x} - \delta \gamma u_{4,x})u_{1,x} + \delta(\gamma u_{3,x} + \beta u_{4,x})u_{2,x} + 3(\beta u_3^2 - 2\delta \gamma u_3 u_4 - \delta \beta u_4^2)u_1^2 \\ \quad + 6\delta(\gamma u_3^2 + 2\beta u_3 u_4 - \delta \gamma u_4^2)u_1 u_2 - 3\delta(\beta u_3^2 - 2\delta \gamma u_3 u_4 - \delta \beta u_4^2)u_2^2], \\ f^{[4]} = \frac{1}{\alpha^4 \delta^2} [-(\gamma u_3 + \beta u_4)u_{1,xx} + (\beta u_3 - \delta \gamma u_4)u_{2,xx} - (\gamma u_1 - \beta u_2)u_{3,xx} - (\beta u_1 + \delta \gamma u_2)u_{4,xx} \\ \quad + (\gamma u_{3,x} + \beta u_{4,x})u_{1,x} - (\beta u_{3,x} - \delta \gamma u_{4,x})u_{2,x} + 3(\gamma u_3^2 + 2\beta u_3 u_4 - \delta \gamma u_4^2)u_1^2 \\ \quad - 6(\beta u_3^2 - 2\delta \gamma u_3 u_4 - \delta \beta u_4^2)u_1 u_2 - 3\delta(\gamma u_3^2 - 2\beta u_3 u_4 - \delta \gamma u_4^2)u_2^2]. \end{cases}$$

Based on these results, we can impose  $\Delta_r = 0$ ,  $m \geq 0$ , to introduce

$$\varphi_{t_m} = \mathcal{N}^{[m]} \varphi = \mathcal{N}^{[m]}(u, \lambda) \varphi, \mathcal{N}^{[m]} = (\lambda^m Y)_+ = \sum_{n=0}^m \lambda^n Y^{[m-n]}, m \geq 0, \quad (2.16)$$

which are the temporal matrix spectral problems within the zero curvature formulation. The conditions that guarantee the solvability of the spatial and temporal matrix spectral problems in (2.5) and (2.16) are the zero curvature equations in (1.6). They lead to a hierarchy of integrable models with four potentials:

$$\begin{aligned} u_{t_m} &= X^{[m]} = X^{[m]}(u) \\ &= (\alpha \delta e^{[m+1]}, -\alpha b^{[m+1]}, \alpha \delta g^{[m+1]}, -\alpha c^{[m+1]})^T, m \geq 0, \end{aligned} \quad (2.17)$$

or more concretely,

$$\begin{aligned} u_{1,t_m} &= \alpha \delta e^{[m+1]}, u_{2,t_m} = -\alpha b^{[m+1]}, \\ u_{3,t_m} &= \alpha \delta g^{[m+1]}, u_{4,t_m} = -\alpha c^{[m+1]}, m \geq 0. \end{aligned} \quad (2.18)$$

We can compute some particular examples. The first nonlinear example is the model of combined integrable nonlinear Schrödinger equations:

$$\begin{cases} u_{1,t_2} = \frac{1}{\alpha^2\delta}[-\beta u_{1,xx} - \delta\gamma u_{2,xx} + 2(\beta u_3 - \delta\gamma u_4)(u_1^2 - \delta u_2^2) + 4\delta(\gamma u_3 + \beta u_4)u_1u_2], \\ u_{2,t_2} = \frac{1}{\alpha^2\delta}[\gamma u_{1,xx} + \beta u_{2,xx} - 2(\gamma u_3 + \beta u_4)(u_1^2 - \delta u_2^2) + 4(\beta u_3 - \delta\gamma u_4)u_1u_2], \\ u_{3,t_2} = \frac{1}{\alpha^2\delta}[\beta u_{3,xx} - \delta\gamma u_{4,xx} - 2(\beta u_1 + \delta\gamma u_2)(u_3^2 - \delta u_4^2) + 4\delta(\gamma u_1 - \beta u_2)u_3u_4], \\ u_{4,t_2} = \frac{1}{\alpha^2\delta}[\gamma u_{3,xx} + \beta u_{4,xx} - 2(\gamma u_1 - \beta u_2)(u_3^2 - \delta u_4^2) + 4(\beta u_1 + \delta\gamma u_2)u_3u_4], \end{cases} \quad (2.19)$$

and the second one is the model of combined integrable modified Korteweg–de Vries equations:

$$\begin{cases} u_{1,t_3} = \frac{1}{\alpha^3\delta}\{-\gamma u_{1,xxx} + \beta u_{2,xxx} + 6[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta\gamma u_4)u_2]u_{1,x} \\ \quad - 6[(\beta u_3 - \delta\gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{2,x}\}, \\ u_{2,t_3} = \frac{1}{\alpha^3\delta^2}\{-\beta u_{1,xxx} - \delta\gamma u_{2,xxx} + 6[(\beta u_3 - \delta\gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{1,x} \\ \quad + 6\delta[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta\gamma u_4)u_2]u_{2,x}\}, \\ u_{3,t_3} = -\frac{1}{\alpha^3\delta}\{-\gamma u_{3,xxx} - \beta u_{4,xxx} + 6[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta\gamma u_4)u_2]u_{3,x} \\ \quad + 6[(\beta u_3 - \delta\gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{4,x}\}, \\ u_{4,t_3} = \frac{1}{\alpha^3\delta^2}\{\beta u_{3,xxx} - \delta\gamma u_{4,xxx} - 6[(\beta u_3 - \delta\gamma u_4)u_1 + \delta(\gamma u_3 + \beta u_4)u_2]u_{3,x} \\ \quad + 6\delta[(\gamma u_3 + \beta u_4)u_1 - (\beta u_3 - \delta\gamma u_4)u_2]u_{4,x}\}. \end{cases} \quad (2.20)$$

These provide two typical coupled integrable models, which extend the category of coupled integrable models of nonlinear Schrödinger equations and modified Korteweg–de Vries equations presented recently (see, e.g. [24, 29–31]). One interesting character is that every equation contains two derivative terms of the highest order, and so, we call them combined models.

Two special cases of  $\beta = 1, \gamma = 0$  and  $\beta = 0, \gamma = 1$  in the obtained hierarchy are interesting and produce reduced hierarchies of uncombined integrable models.

If we take  $\alpha = \delta = 1, \beta = 1$  and  $\gamma = 0$  in the model (2.19), we obtain a coupled integrable nonlinear Schrödinger type model:

$$\begin{cases} u_{1,t_2} = -u_{1,xx} + 2u_3(u_1^2 - u_2^2) + 4u_1u_2u_4, \\ u_{2,t_2} = -u_{2,xx} - 2u_4(u_1^2 - u_2^2) + 4u_1u_2u_3, \\ u_{3,t_2} = u_{3,xx} - 2u_1(u_3^2 - u_4^2) - 4u_2u_3u_4, \\ u_{4,t_2} = u_{4,xx} + 2u_2(u_3^2 - u_4^2) - 4u_1u_3u_4. \end{cases} \quad (2.21)$$

If we take  $\alpha = \delta = 1, \beta = 0$  and  $\gamma = 1$  in the model (2.19), we obtain another coupled integrable nonlinear Schrödinger type model:

$$\begin{cases} u_{1,t_2} = -u_{2,xx} - 2u_4(u_1^2 - u_2^2) + 4u_1u_2u_3, \\ u_{2,t_2} = u_{1,xx} - 2u_3(u_1^2 - u_2^2) - 4u_1u_2u_4, \\ u_{3,t_2} = -u_{4,xx} - 2u_2(u_3^2 - u_4^2) + 4u_1u_3u_4, \\ u_{4,t_2} = u_{3,xx} - 2u_1(u_3^2 - u_4^2) - 4u_2u_3u_4. \end{cases} \quad (2.22)$$

If we take  $\alpha = \delta = 1, \beta = 1$  and  $\gamma = 0$  in the model (2.20), we obtain a coupled integrable modified Korteweg–de Vries type

model:

$$\begin{cases} u_{1,t_3} = u_{2,xxx} + 6(u_1u_4 - u_2u_3)u_{1,x} - 6(u_1u_3 + u_2u_4)u_{2,x}, \\ u_{2,t_3} = -u_{1,xxx} + 6(u_1u_3 + u_2u_4)u_{1,x} + 6(u_1u_4 - u_2u_3)u_{2,x}, \\ u_{3,t_3} = -u_{4,xxx} + 6(u_1u_4 - u_2u_3)u_{3,x} + 6(u_1u_3 + u_2u_4)u_{4,x}, \\ u_{4,t_3} = u_{3,xxx} - 6(u_1u_3 + u_2u_4)u_{3,x} + 6(u_1u_4 - u_2u_3)u_{4,x}. \end{cases} \quad (2.23)$$

If we take  $\alpha = \delta = 1, \beta = 0$  and  $\gamma = 1$  in the model (2.20), we obtain another coupled integrable modified Korteweg–de Vries type model:

$$\begin{cases} u_{1,t_3} = -u_{1,xxx} + 6(u_1u_3 + u_2u_4)u_{1,x} + 6(u_1u_4 - u_2u_3)u_{2,x}, \\ u_{2,t_3} = -u_{2,xxx} - 6(u_1u_4 - u_2u_3)u_{1,x} + 6(u_1u_3 + u_2u_4)u_{2,x}, \\ u_{3,t_3} = -u_{3,xxx} + 6(u_1u_3 + u_2u_4)u_{3,x} - 6(u_1u_4 - u_2u_3)u_{4,x}, \\ u_{4,t_3} = -u_{4,xxx} + 6(u_1u_4 - u_2u_3)u_{3,x} + 6(u_1u_3 + u_2u_4)u_{4,x}. \end{cases} \quad (2.24)$$

These models are different from the vector AKNS integrable models [28]. In each pair, the two models just exchange the first component with the second component and the third component with the fourth component, plus a sign change for one of the two components, in the vector fields on the right hand sides. Moreover, all those four models still commute with each other.

### 3. Recursion operator and bi-Hamiltonian formulation

To present a bi-Hamiltonian formulation and show the Liouville integrability for the soliton hierarchy (2.18), one can make use of the so-called trace identity (1.8) associated with the spatial matrix spectral problem (2.5). Plugging the spectral matrix  $\mathcal{M}$  by (2.5) and the Laurent series solution  $Y$  determined by (2.6) into the trace identity leads to

$$-\frac{\delta}{\delta u} \int \lambda^{-(n+1)} \alpha \delta f^{[n+1]} dx = \lambda^{-\kappa} \frac{\partial}{\partial \lambda} \lambda^{\kappa-n} (c^{[n]}, \delta g^{[n]}, b^{[n]}, \delta e^{[n]})^T, \quad n \geq 0, \quad (3.1)$$

where we have used

$$\text{tr}\left(Y \frac{\partial \mathcal{M}}{\partial \lambda}\right) = -2\alpha \delta f, \quad \text{tr}\left(Y \frac{\partial \mathcal{M}}{\partial u}\right) = (2c, 2\delta g, 2b, 2\delta e)^T. \quad (3.2)$$

Checking with  $n = 2$  leads to  $\kappa = 0$ , and accordingly, one arrives at

$$\frac{\delta}{\delta u} \mathcal{H}^{[n]} = (c^{[n+1]}, \delta g^{[n+1]}, b^{[n+1]}, \delta e^{[n+1]})^T, \quad n \geq 0, \quad (3.3)$$

where the Hamiltonian functionals are computed as follows:

$$\mathcal{H}^{[n]} = \int \frac{\alpha \delta}{n+1} f^{[n+2]} dx, \quad n \geq 0. \quad (3.4)$$

This enables us to get a Hamiltonian formulation for the hierarchy (2.18):

$$u_{t_m} = X^{[m]} = J_1 \frac{\delta \mathcal{H}^{[m]}}{\delta u}, \quad m \geq 0, \quad (3.5)$$

where  $J_1$  is the Hamiltonian operator:

$$J_1 = \left[ \begin{array}{c|cc} 0 & 0 & \alpha \\ \hline 0 & \alpha & 0 \\ -\alpha & 0 & 0 \end{array} \right], \quad (3.6)$$

and  $\mathcal{H}^{[m]}$  are the functionals defined by (3.4). Now from the Hamiltonian theory, we see that there exists an interrelation  $S = J_1 \frac{\delta \mathcal{H}}{\delta u}$  between a symmetry  $S$  and a conserved functional  $\mathcal{H}$  of the same model.

It is a common fact that these vector fields  $X^{[n]}$  constitutes an Abelian algebra:

$$\begin{aligned} \llbracket X^{[n_1]}, X^{[n_2]} \rrbracket &= X^{[n_1]'}(u)[X^{[n_2]}] - X^{[n_2]'}(u)[X^{[n_1]}] \\ &= 0, \quad n_1, n_2 \geq 0, \end{aligned} \quad (3.7)$$

which is a consequence of an Abelian algebra of Lax operators:

$$\begin{aligned} \llbracket \mathcal{N}^{[n_1]}, \mathcal{N}^{[n_2]} \rrbracket \\ = \mathcal{N}^{[n_1]'}(u)[X^{[n_2]}] - \mathcal{N}^{[n_2]'}(u)[X^{[n_1]}] + [\mathcal{N}^{[n_1]}, \mathcal{N}^{[n_2]}] = 0, \quad n_1, n_2 \geq 0. \end{aligned} \quad (3.8)$$

More discussions about the isospectral zero curvature equations is given in [34].

On the other hand, directly from the recursion relation  $X^{[m+1]} = \Phi X^{[m]}$ , we can compute a hereditary recursion operator  $\Phi = (\Phi_{jk})_{4 \times 4}$  [32] for the hierarchy (2.18) as follows:

$$\begin{cases} \Phi_{11} = \frac{1}{\alpha}(-2u_1 \partial^{-1}u_4 + 2u_2 \partial^{-1}u_3), \quad \Phi_{12} = \frac{1}{\alpha}(-\partial_x + 2u_1 \partial^{-1}u_3 + 2\delta u_2 \partial^{-1}u_4, \\ \Phi_{13} = \frac{1}{\alpha}(2u_1 \partial^{-1}u_2 + 2u_2 \partial^{-1}u_1), \quad \Phi_{14} = \frac{1}{\alpha}(-2u_1 \partial^{-1}u_1 + 2\delta u_2 \partial^{-1}u_2); \end{cases} \quad (3.9)$$

$$\begin{cases} \Phi_{21} = \frac{1}{\alpha}(\frac{1}{\delta} \partial_x - \frac{2}{\delta} u_1 \partial^{-1}u_3 - 2u_2 \partial^{-1}u_4), \quad \Phi_{22} = \frac{1}{\alpha}(-2u_1 \partial^{-1}u_4 + 2u_2 \partial^{-1}u_3), \\ \Phi_{23} = \frac{1}{\alpha}(-\frac{1}{\delta} \partial_x - \frac{2}{\delta} u_1 \partial^{-1}u_1 + 2u_2 \partial^{-1}u_2), \quad \Phi_{24} = \frac{1}{\alpha}(-2u_1 \partial^{-1}u_2 - 2u_2 \partial^{-1}u_1); \end{cases} \quad (3.10)$$

$$\begin{cases} \Phi_{31} = \frac{1}{\alpha}(2u_3 \partial^{-1}u_4 + 2u_4 \partial^{-1}u_3), \quad \Phi_{32} = \frac{1}{\alpha}(-2u_3 \partial^{-1}u_3 + 2\delta u_4 \partial^{-1}u_4), \\ \Phi_{33} = \frac{1}{\alpha}(-2u_3 \partial^{-1}u_2 + 2u_4 \partial^{-1}u_1), \quad \Phi_{34} = \frac{1}{\alpha}(-\partial_x + 2u_3 \partial^{-1}u_1 + 2\delta u_4 \partial^{-1}u_2); \end{cases} \quad (3.11)$$

$$\begin{cases} \Phi_{41} = \frac{1}{\alpha}(-\frac{2}{\delta} u_3 \partial^{-1}u_3 + 2u_4 \partial^{-1}u_4), \quad \Phi_{42} = \frac{1}{\alpha}(-2u_3 \partial^{-1}u_4 - 2u_4 \partial^{-1}u_3), \\ \Phi_{43} = \frac{1}{\alpha}(\frac{1}{\delta} \partial_x - \frac{2}{\delta} u_3 \partial^{-1}u_1 - 2u_4 \partial^{-1}u_2), \quad \Phi_{44} = \frac{1}{\alpha}(-2u_3 \partial^{-1}u_2 + 2u_4 \partial^{-1}u_1). \end{cases} \quad (3.12)$$

With some analysis, we can see that  $J_1$  and  $J_2 = \Phi J_1$  constitute a Hamiltonian pair. This means that an arbitrary linear combination of  $J_1$  and  $J_2$  is again Hamiltonian, and thus the hierarchy (2.18) possesses a bi-Hamiltonian formulation [33]:

$$u_{t_m} = X^{[m]} = J_1 \frac{\delta \mathcal{H}^{[m]}}{\delta u} = J_2 \frac{\delta \mathcal{H}^{[m-1]}}{\delta u}, \quad m \geq 1. \quad (3.13)$$

Moreover, we can observe that the associated Hamiltonian functionals commute with each other under the corresponding two Poisson brackets [7]:

$$\{\mathcal{H}^{[n_1]}, \mathcal{H}^{[n_2]}\}_{J_1} = \int \left( \frac{\delta \mathcal{H}^{[n_1]}}{\delta u} \right)^T J_1 \frac{\delta \mathcal{H}^{[n_2]}}{\delta u} dx = 0, \quad n_1, n_2 \geq 0, \quad (3.14)$$

and

$$\{\mathcal{H}^{[n_1]}, \mathcal{H}^{[n_2]}\}_{J_2} = \int \left( \frac{\delta \mathcal{H}^{[n_1]}}{\delta p} \right)^T J_2 \frac{\delta \mathcal{H}^{[n_2]}}{\delta u} dx = 0, \quad n_1, n_2 \geq 0. \quad (3.15)$$

To summarize, each model in the hierarchy (2.18) is Liouville integrable and possesses infinitely many commuting symmetries  $\{X^{[n]}\}_{n=0}^{\infty}$  and conserved functionals  $\{\mathcal{H}^{[n]}\}_{n=0}^{\infty}$ . Two specific examples of such nonlinear combined Liouville integrable Hamiltonian models are the two models in (2.19) and (2.20)

#### 4. Concluding remarks

A Liouville integrable hierarchy with four potentials has been generated from a newly introduced specific special matrix spectral problem, along with their bi-Hamiltonian formulations. The success comes from the existence of a particular Laurent series solution of the corresponding stationary zero curvature equation. The resulting integrable models involve two arbitrary constants,  $\beta$  and  $\gamma$ , and one arbitrary nonzero constant,  $\delta$ , and thus, they contain diverse specific four-component examples of integrable models.

It should be particularly interesting to study algebraic or geometric structures of soliton solutions to the obtained integrable models. One can try to use different powerful and effective approaches, such as the Riemann-Hilbert technique [35], the Darboux transformation [36–39], the Zakharov–Shabat dressing method [40], the algebro-geometric method [41–45], the decomposition method [46–49], and the determinant approach [50]. Besides solitons, lump, kink, breather and rogue wave solutions, including their interaction solutions (see, e.g. [51–59]), are also of interest, and one can compute them from solitons by taking wave number reductions. Additionally, conducting nonlocal group reductions or similarity transformations for matrix spectral problems, one can get nonlocal reduced integrable models as well as study their soliton solutions (see, e.g. [60–62]).

Integrable models are of great interest. There is a huge diversity of multi-component integrable models, which have close connections to various areas of mathematics, including algebraic geometry, Lie theory, and the theory of special functions. Studies on structures of multi-component integrable models will advance our understanding of complex nonlinear mathematical and physical problems.

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